

Text as Data: Computational Methods for Social Science Research

文本作為資料：社會科學研究的計算方法

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Room:	國發 204	Credits:	2
Audience:	Postgraduate	Time:	Fri. 16:30–18:20
Semester:	115-1 Fall	Duration:	16 weeks
Format:	Lecture and take-home labs	Office Hours:	Instructor / TA
Pre-course Survey:	Complete the survey	Course Page:	GitHub

Acknowledgments: The design of this course draws primarily on Stefan Müller’s quantitative text analysis syllabus. The lab exercises and teaching materials are adapted from those developed by Martijn Schoonvelde for the 2026 Essex Summer School course “Introduction to QTA.”

1 Course Description

Text-as-data and computational text analysis has become increasingly prominent in political science (Grimmer and Stewart 2013; Wilkerson and Casas 2017; Gentzkow, Kelly, and Taddy 2019), for a simple reason: most of what politicians, parties, and citizens do leaves a digital text record. Legislative corpora like ParlSpeech V2 (Rauh and Schwalbach 2020), the Manifesto Project Database (Lehmann et al. 2025), and legisTaiwan (Liao, Tang, and OpenFund Ltd. 2025) make it possible to study political behavior at a scale that wasn’t feasible a decade ago. Designed for graduate students in political science and the social sciences, the course treats computational methods as tools for answering substantive research questions—about representation, polarization, party competition, and political communication—rather than as ends in themselves. The module covers both quantitative text analysis and NLP, with attention to underlying assumptions, applications, and implementation in R and Python.

We begin with the assumptions behind text-as-data research, tokenisation, and document-feature matrices, then connect these representations to dictionaries and sentiment analysis. Subsequent sessions introduce human coding, classifiers such as Naive Bayes and SVM, text scaling, and topic models, including LDA, STM, and Keyword-Assisted Topic Models (Eshima, Imai, and Sasaki 2024). Each method is paired with published applications from political science, including work that measures party positions, tracks issue attention, and detects conflict in legislative speech. Students examine what kinds of inferences a technique can support and where those inferences break down. These approaches provide baselines for later comparisons with embeddings and transformer-based models.

Week 9 connects readability and text similarity to word embeddings. Students learn the intuition behind Word2Vec (Mikolov et al. 2013), GloVe (Pennington, Socher, and Manning 2014), and FastText (Bojanowski et al. 2017), then examine applications of context-specific embeddings (Rodriguez, Spirling, and Stewart 2023a; Wirsching et al. 2025): how does the meaning of “immigration” vary across speakers, periods, or languages? Guided examples use supplied models; extended embedding-regression and multilingual workflows, including conText, are optional extensions.

Week 10 links classical NLP tasks—POS tagging, named entity recognition, and dependency parsing—to contextual representations and transformer-based classification. BERT (Devlin et al. 2019), RoBERTa (Liu et al. 2019), and DistilBERT (Sanh et al. 2019b) illustrate how pre-trained models can be adapted to social-science tasks. The emphasis is on conceptual understanding, published applications, comparison with simpler baselines, and validation. Fine-tuning and interpretability are introduced through pre-run demonstrations; complete training workflows and SHAP/LIME implementations are supplementary. Week 12 examines prompting and evaluating large language models for annotation, classification, and extraction, using hosted or local models. Multi-agent systems such as AutoGen (Wu et al. 2023) are discussed as an extension, with attention to the limits of delegating research tasks to models.

Take-home labs complement lectures and discussion. Students work through guided exercises independently outside class, following the text-analysis workflow: define the research question and corpus, preprocess the text, construct an appropriate representation, apply a method, and validate the interpretation. Weeks 2 and 4 introduce supplied political-text corpora and basic checks on data availability, sampling, metadata, and access conditions so that students can develop feasible projects before the Week 7 abstract deadline. Week 13 explores new forms of textual data, including multilingual texts, speech-to-text, and multimodal sources, asking how translation, transcription, and model-generated annotations affect social-science findings. By the end of the course, students should be able to select and evaluate methods in relation to substantive research questions.

2 Course Objectives

Upon successful completion of the course, students will be able to:

1. Apply core text analysis methods spanning classical NLP (tokenization, POS tagging, NER) and modern representations (embeddings, transformer-based models), with a working grasp of reliability, validation, and accuracy as methodological concerns.
2. Build supervised classifiers and use large language models through prompt engineering and few-shot learning—knowing, in a given task, when traditional machine learning is the better choice over a pre-trained LLM.
3. Carry out an original text analysis research project that addresses a substantive question in political science, attentive to reproducibility and methodological transparency throughout.
4. Read published text-analysis research with a critical eye: is the method rigorous, are the inferences sound, and was the right technique chosen for the job?

Note: This is an introductory course designed for students in the Graduate Institute of National Development and related social science disciplines. No advanced technical training is required. Instead, the course emphasizes how text-as-data and NLP techniques can be applied to answer substantive research questions in political science, political communication, legal studies, sociology, and other social science fields.

3 Course Requirements

This course requires prior experience with R and comfort running some code in a Python environment. Students will need basic programming skills to implement text analysis methods, complete assignments, and work with real political and social science text data. Students should also be comfortable using an IDE (RStudio or Positron). GitHub is the primary source for course materials and code. Written coursework and posters are submitted through shared Overleaf projects. A setup guide will be provided before Week 2. Week 1 introduces the course rules, expectations, and assignment workflow; GitHub Desktop and IDE setup will be covered in Week 2.

- **Programming Languages:** The course is taught primarily in R. Most lectures, examples, and methods use R packages like `quanteda`, `tidytext`, `text2vec`, `udpipe`, `keyATM`, and `spacyr`. Python is introduced selectively for transformer models (`transformers`, `flair`), NLP toolkits (`gensim`, `spacy`), and topic modeling (`BERTopic`).
- **Development Environment:** Students will work primarily in RStudio (for R) or Positron (for Python and cross-language work). Essential skills include editing and running code, debugging, managing packages, and using integrated terminals. Familiarity with R Markdown / Quarto

(RStudio) and Jupyter / Quarto notebooks (Positron) is helpful but not required. We will cover installation, project setup, and the practical workflow in Week 2. Students should review the setup guide beforehand; no software installation or coding exercise is required during Week 1.

No Git experience required. We'll use GitHub Desktop (<https://desktop.github.com>), a point-and-click interface that handles version control without command-line work. RStudio and Positron also have built-in Git tools. Week 1 explains the rules of the course and how materials, assignments, and communication are organized. Week 2 introduces accessing course materials, cloning repositories, and saving changes to practice code; students practise these steps in the take-home lab.

Recommended: Apply for the free GitHub Student Developer Pack (<https://education.github.com/pack>) using your university email. This gives you GitHub Pro, Copilot, and other developer tools at no cost.

- **What Students Will Actually Do:** Labs are designed primarily as self-directed take-home practice. Students access guided exercises from the course GitHub repository and work through them outside class. They write code to preprocess text data, implement analysis methods, create visualizations, and interpret results. Students can book instructor or TA office hours for help with concepts, code, or troubleshooting. Written coursework and posters are prepared in $\text{\LaTeX}$ using course templates and submitted through Overleaf with instructor editing access. Students check that their projects compile to PDF before submission. Practical work develops debugging, package-management, and reproducible-workflow skills.

Beginners should expect to spend extra time on the technical side—learning to read error messages, troubleshoot package conflicts, and structure reproducible analyses. These skills are as important as understanding the methods themselves.

- **Optional Exploration:** We may use LM Studio (<https://lmstudio.ai>) to run open-source LLMs locally. This lets students experiment with LLMs like LLaMA and Mistral without API costs. Installation instructions will be provided if we use it.

4 Student Workload (Expected weekly study hours before and/or after class)

This is a 2-credit course introducing NLP and text-as-data methods through concepts, published social-science applications, discussion, and guided practice. Class time focuses on lectures, discussion, and demonstrations; labs are primarily completed independently at home from Week 2 onward. Plan on approximately **3–4 hours per week** outside class, including take-home labs, readings, and steady work on the research project:

Activity	Hours / week
<i>Before class:</i> Reading the assigned text-as-data papers and reviewing the week's slides	1–2
<i>After class:</i> Working through take-home labs, reviewing code, and checking results	0.5–1
<i>Research project:</i> Developing the abstract, research design, analysis, poster, and final paper	1.5–2
Typical weekly total (outside class)	3–4

Project work should begin early and continue throughout the semester. More time may be needed near submission deadlines or when learning unfamiliar software. The estimate above does not include optional supplementary readings or advanced exercises.

Note on the readings: we examine each paper's research question, data, measurement choices, substantive findings, and limitations. The aim is to understand how a method supports a social-science argument, not to reproduce every technical detail.

5 Course Readings

5.1 Reading Policy

This course does not require students to purchase any textbook. Each regular teaching week assigns **three core academic readings**, primarily journal articles. An assigned book chapter counts as one reading, as does a journal article about software. **Technical documentation and tutorials are listed separately and do not count toward this limit**; consult the relevant sections for setup and take-home labs. Supplementary readings are optional, not additional required preparation. No new core readings are assigned for holidays, the poster workshop, or the final poster presentations.

Selected book chapters may be assigned when they provide necessary conceptual or technical scaffolding. Students are not expected to buy these books; required excerpts from books or edited volumes will be made available by the instructor or through library-access channels. Copyright-restricted readings will not be posted on public GitHub repositories. Where slides draw on copyrighted material, the relevant excerpts are included under fair-use / copyright-compliant terms and are for the use of enrolled students only.

5.2 General Readings and Textbook References

The works below provide general background for the course. Selected chapters may also appear in the weekly readings when they are especially useful for the course topic.

Quantitative text analysis and NLP

- Van Atteveldt, Wouter, Trilling, Damian, and Arcila Calderón, Carlos (2022). *Computational Analysis of Communication*. Wiley Blackwell. URL: <https://cssbook.net/>. Free online textbook; no purchase required. Updated online draft: <https://v2.cssbook.net/>.
- Nielbo, Kristoffer L et al. (2024). “Quantitative Text Analysis”. *Nature Reviews Methods Primers* 4(1): 25. DOI: [10.1038/s43586-024-00302-w](https://doi.org/10.1038/s43586-024-00302-w)
- Benoit, Kenneth (2020). “Text as Data: An Overview”. In: *Handbook of Research Methods in Political Science and International Relations*. Ed. by Curini, Luigi and Franzese, Robert. Thousand Oaks: Sage: 461–497. DOI: [10.4135/9781526486387.n29](https://doi.org/10.4135/9781526486387.n29)
- Watanabe, Kohei and Müller, Stefan (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io> (visited on 01/01/2024)
- Stoltz, Dustin S. and Taylor, Marshall A. (2024). *Mapping Texts: Computational Text Analysis for the Social Sciences*. Oxford: Oxford University Press. DOI: [10.1093/oso/9780197756874.001.0001](https://doi.org/10.1093/oso/9780197756874.001.0001)
- Hvitfeldt, Emil and Silge, Julia (2021). *Supervised Machine Learning for Text Analysis in R*. CRC Press. URL: <https://smltar.com/>
- Grimmer, Justin, Roberts, Margaret E., and Stewart, Brandon M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691207544/text-as-data>
- Jurafsky, Daniel and Martin, James H. (2026). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd. Online manuscript released August 19, 2026. URL: <https://web.stanford.edu/~jurafsky/slp3/>

R, Quarto, data science, and research design

- Wickham, Hadley, Çetinkaya-Rundel, Mine, and Grolemund, Garrett (2023). *R for Data Science (2nd Edition)*. O’Reilly Media. URL: <https://r4ds.hadley.nz/>
- Alexander, Rohan (2023). *Telling Stories with Data: With Applications in R*. New York: CRC Press. URL: <https://tellingstorieswithdata.com>
- Huntington-Klein, Nick (2025). *The Effect: An Introduction to Research Design and Causality*. 2nd ed. Chapman and Hall/CRC. URL: <https://www.theeffectbook.net/>

- Cunningham, Scott (2021). *Causal Inference: The Mixtape*. Yale University Press. URL: <https://www.scunning.com/mixtape>
- Turrell, Arthur (2024). *Coding for Economists*. URL: <https://aeturrell.github.io/coding-for-economists/intro.html>
- Healy, Kieran (2019). *Data Visualization: A Practical Introduction*. Princeton: Princeton University Press. URL: <https://socviz.co>
- Tunstall, Lewis, Werra, Leandro von, and Wolf, Thomas (2022). *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*. O'Reilly Media. URL: <https://www.oreilly.com/library/view/natural-language-processing/9781098136789/>

6 Office Hours

Instructor student hours: [Book an appointment.](#)

TA office hours: [Book an appointment.](#)

Students are welcome to book instructor or TA office hours for help with take-home labs, including questions about methods, code, errors, and interpretation. Bring the relevant code or output and a short description of where you need help.

Getting help:

- *Course technical questions:* GitHub Issues may be used for non-sensitive questions about the course code. Send personal matters and questions about individual grades directly to the instructor or TA.
- *Software/package problems:* I'll help you learn to raise issues on GitHub with package authors.
- *Other questions:* Email me with a short description, and I'll arrange a time to meet.
- **Your wellbeing:** *If you're feeling stressed or overwhelmed, please don't tough it out alone—email me or come talk to me after class, and I'll do my best to be flexible with deadlines. If you'd like to talk to a professional, NTU offers the following resources:*
 - **Student Counseling Center** (an initial intake appointment is required; walk-ins are welcome in urgent situations): <https://my.ntu.edu.tw/counsel/index.html> | (02) 3366-2181. Main Campus: Wangle Building (望樂樓), next to Graduate Dorm 1 on Zhoushan Rd. (Gongguan MRT, Exit 2); City Campus: 3F, College of Medicine Library.
 - **In-college student support** (a counselor based in your own college; walk-ins welcome): <https://csw.ntu.edu.tw/>

- **Emergencies:** Campus Security Center, 24 hours: (02) 3366-9119. If life is in danger, call **119** (medical) or **110** (police) right away.
- **Free national hotlines (24 hours):** Mental Health Care Line **1925**; Lifeline **1995**.

7 Course Schedule

7.1 Course Content Summary

Week 1: Introduction to Quantitative Text Analysis

Introduces the basic concepts of quantitative text analysis and natural language processing, outlines course structure and expectations, and surveys the text-as-data paradigm in social science research.

Week 2: Text Preparation in R and Python: Collaborative Workflow

Covers the use of R and Python for text analysis, sets up development environments (RStudio and Positron), and establishes a GitHub-based coding workflow with written coursework submitted through shared Overleaf projects. Introduces supplied corpora and a basic data-feasibility checklist for students' research projects.

Week 3: Mid-Autumn Festival (No Class)

No class on September 25.

Week 4: Tokenisation, Document-Feature Matrices, Dictionaries, and Sentiment Analysis

Connects preprocessing decisions and bag-of-words representations to dictionary-based measurement. A single workflow moves from a documented corpus to tokens, document-feature matrices, sentiment scores, and validation in political texts. Students also check corpus boundaries and access conditions for potential projects.

Week 5: National Day Observed (No Class)

No class on October 9.

Week 6: Human Coding and Document Classification

Covers human coding, training-set construction, supervised document classification, validation, and performance metrics. Students connect coding rules and annotation quality to the substantive claims a classifier can support.

Week 7: Supervised, Unsupervised, and Semi-Supervised Scaling

Introduces Wordscores, Wordfish, and Latent Semantic Scaling (LSS) for estimating positions from text, with attention to anchors, assumptions, and external validation. **Individual research abstract due: October 23.**

Week 8: Topic Models and Their Variants

Covers LDA, structural topic models, and keyword-assisted topic models, with emphasis on interpreting topics, comparing issue attention, and validating topics as social-science measures.

Week 9: Describing and Comparing Texts: Readability, Similarity, and Word Embeddings

Combines readability and complexity measures with count-based similarity, word embeddings, and context-specific meaning. Students compare how different representations support research on political language across actors and groups.

Week 10: NLP Pipelines and Transformers: Annotation, Classification, and Interpretability

Connects POS tagging, named entity recognition, and dependency parsing to contextual representations and transformer-based classification. Fine-tuning and interpretability are introduced through applications and demonstrations, with detailed implementation reserved for supplementary materials.

Week 11: NTU Sports Day (No Class)

No class on November 20.

Week 12: Large Language Models and Multi-Agent Systems

Introduces LLMs for annotation, classification, summarization, and multi-agent workflows. Applications foreground reproducibility, validation, bias, and the limits of substituting model outputs or simulations for human evidence.

Week 13: New Forms of Textual Data: Multilingual and Multimodal Sources

Explores research with multilingual texts, political speech recordings, and audiovisual sources. Students examine how machine translation, speech recognition, and model-generated annotations change the evidence available for political and social research, using published applications and a guided comparison of original and transformed text. **Midterm research design report due: December 4.**

Week 14: Poster Workshop: Format, Visual Presentation, and 5-Minute Talk

This session focuses on poster format, visual presentation, and the 5-minute talk. We will discuss revision directions collectively as preparation for the formal poster presentations in Week 15. Students are encouraged to share a draft through Overleaf before the formal presentation and discuss revisions with the instructor after class or during office hours.

Week 15: Final Poster Presentations

The formally assessed final poster presentations take place on December 18. Students present research posters featuring their analysis and results, give approximately 5-minute introductions, and respond to questions and feedback from faculty, classmates, and other participants.

Week 16: Constitution Day Holiday (No Class)

No class on December 25. Students revise their projects independently after the poster session.

Manuscript due: TBA. Bonus claim deadline: TBA. Both deadlines will be announced on NTU COOL. Submit the final research paper through Overleaf with instructor editing access; include any bonus claim and its supporting documents in a shared Overleaf project.

Week	Date	Topic	Notes
W1	Sep 11	Introduction to Quantitative Text Analysis	
W2	Sep 18	Text Preparation in R and Python: Collaborative Workflow	
W3	Sep 25	Mid-Autumn Festival (No Class)	
W4	Oct 2	Tokenisation, Document-Feature Matrices, Dictionaries, and Sentiment Analysis	
W5	Oct 9	National Day Observed (No Class)	
W6	Oct 16	Human Coding and Document Classification	
W7	Oct 23	Supervised, Unsupervised, and Semi-Supervised Scaling	Individual research abstract due
W8	Oct 30	Topic Models and Their Variants	
W9	Nov 6	Describing and Comparing Texts: Readability, Similarity, and Word Embeddings	
W10	Nov 13	NLP Pipelines and Transformers: Annotation, Classification, and Interpretability	
W11	Nov 20	NTU Sports Day (No Class)	
W12	Nov 27	Large Language Models and Multi-Agent Systems	
W13	Dec 4	New Forms of Textual Data: Multilingual and Multimodal Sources	Midterm research design report due
W14	Dec 11	Poster Workshop: Format, Visual Presentation, and 5-Minute Talk	Poster revision clinic
W15	Dec 18	Final Poster Presentations	5-minute talks; peer feedback
W16	Dec 25	Constitution Day Holiday (No Class)	Manuscript due: TBA; bonus deadline: TBA

7.2 Weekly Schedule

Only the three items under **Core Reading** are required academic preparation for each regular teaching week. Technical documents support practical work; all supplementary readings are optional.

Week 1: Introduction to Quantitative Text Analysis

Topics:

- What are quantitative text analysis and natural language processing?
- Course structure and expectations
- Overview of text-as-data paradigm in social science
- Why measurement validity, context, and language matter when choosing computational methods

Core Reading:

- Benoit, Kenneth (2020). “Text as Data: An Overview”. In: *Handbook of Research Methods in Political Science and International Relations*. Ed. by Curini, Luigi and Franzese, Robert. Thousand Oaks: Sage: 461–497. DOI: [10.4135/9781526486387.n29](https://doi.org/10.4135/9781526486387.n29) (instructor-provided chapter excerpt)
- Grimmer, Justin and Stewart, Brandon M (2013). “Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts”. *Political Analysis* 21(3): 267–297
- Baden, Christian et al. (2022). “Three Gaps in Computational Text Analysis Methods for Social Sciences: A Research Agenda”. *Communication Methods and Measures* 16(1): 1–18. DOI: [10.1080/19312458.2021.2015574](https://doi.org/10.1080/19312458.2021.2015574)

Supplementary Reading:

- Grimmer, Justin, Roberts, Margaret E, and Stewart, Brandon M (2021). “Machine Learning for Social Science: An Agnostic Approach”. *Annual Review of Political Science* 24(1): 395–419
- Wilkerson, John and Casas, Andreu (2017). “Large-scale Computerized Text Analysis in Political Science: Opportunities and Challenges”. *Annual Review of Political Science* 20(1): 529–544
- Gentzkow, Matthew, Kelly, Bryan, and Taddy, Matt (2019). “Text as Data”. *Journal of Economic Literature* 57(3): 535–574
- Nielbo, Kristoffer L et al. (2024). “Quantitative Text Analysis”. *Nature Reviews Methods Primers* 4(1): 25. DOI: [10.1038/s43586-024-00302-w](https://doi.org/10.1038/s43586-024-00302-w)

- Egami, Naoki, Fong, Christian J., et al. (2022). “How to Make Causal Inferences Using Texts”. *Science Advances* 8(42): eabg2652

Course orientation:

- Course expectations, assessment, participation, and AI-use disclosure
- Overview of course tools and learning resources
- Student backgrounds, research interests, and the preparation checklist for Week 2

Software installation, repository setup, and the first take-home lab are introduced in Week 2.

Week 2: Text Preparation in R and Python: Collaborative Workflow

Topics:

- Using R and Python for text analysis
- Setting up development environments: RStudio and Positron
- GitHub Desktop, private code repositories, and Overleaf submissions with instructor editing access
- Structuring workflows for quantitative research projects
- Supplied political-text corpora and a first check of data access, sampling, and project feasibility

Core Reading:

- Grimmer, Justin, Roberts, Margaret E., and Stewart, Brandon M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691207544/text-as-data>, Chapter 4 (instructor-provided excerpt)
- Welbers, Kasper, Van Attevelde, Wouter, and Benoit, Kenneth (2017). “Text Analysis in R”. *Communication Methods and Measures* 11(4): 245–265
- Benoit, Kenneth, Watanabe, Kohei, et al. (2018). “quanteda: An R Package for the Quantitative Analysis of Textual Data”. *Journal of Open Source Software* 3(30): 774

Supplementary Reading:

- Watanabe, Kohei and Müller, Stefan (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io> (visited on 01/01/2024), Chapter 1
- Alexander, Rohan (2023). *Telling Stories with Data: With Applications in R*. New York: CRC Press. URL: <https://tellingstorieswithdata.com>, Chapter 3

- Wickham, Hadley, Çetinkaya-Rundel, Mine, and Grolemund, Garrett (2023). *R for Data Science (2nd Edition)*. O'Reilly Media. URL: <https://r4ds.hadley.nz/>, Chapters 1–2
- Liao, Yen-Chieh, Akbik, Alan, et al. (2025). *flairR: An R Wrapper for Accessing Flair NLP*. R package version 0.1.8. URL: <https://davidycliao.github.io/flairR>
- Manning, Christopher D., Raghavan, Prabhakar, and Schütze, Hinrich (2008). *Introduction to Information Retrieval*. Cambridge University Press. URL: <https://nlp.stanford.edu/IR-book/>, Chapter 2

Technical Documents (not counted toward the reading limit):

- Overleaf: [Learn L^AT_EX in 30 minutes](#); course templates will be provided
- GitHub Desktop Documentation: <https://docs.github.com/en/desktop> (Skim “Getting Started” section)
- Quarto Documentation: <https://quarto.org/docs/get-started/> (Optional, for analysis notebooks only; assessed writing uses L^AT_EX)
- Posit Blog: [Privacy and AI Assistants](#) (Skim)

Take-home lab:

- Installing packages and dependencies; setting up a project in RStudio or Positron
- Creating a GitHub account if needed, setting up GitHub Desktop, and cloning the course repository
- Creating a short L^AT_EX document in Overleaf, compiling it to PDF, and inviting the instructor with Editor access; practising GitHub commits
- Inspecting a supplied corpus and identifying one feasible text source for a possible research project

Week 3: Mid-Autumn Festival (No Class)

No class on September 25.

Week 4: Tokenisation, Document-Feature Matrices, Dictionaries, and Sentiment Analysis

Topics:

- Bag-of-words assumptions; tokens, types, and features; stemming and lemmatisation
- Building document-feature matrices and understanding how preprocessing affects measurement

- Dictionary construction, multi-word expressions, and lexicon-based sentiment analysis
- Political applications: measuring conflict and issue-related language in legislative speech
- Validating dictionary scores through context, human coding, and error analysis

Core Reading:

- Denny, Matthew W. and Spirling, Arthur (2018). “Text Preprocessing For Unsupervised Learning: Why It Matters, When It Misleads, And What To Do About It”. *Political Analysis* 26(2): 168–189
- Rauh, Christian (2018). “Validating a Sentiment Dictionary for German Political Language—A Workbench Note”. *Journal of Information Technology & Politics* 15(4): 319–343. DOI: [10.1080/19331681.2018.1485608](https://doi.org/10.1080/19331681.2018.1485608)
- Proksch, Sven-Oliver, Lowe, Will, et al. (2019). “Multilingual Sentiment Analysis: A New Approach to Measuring Conflict in Legislative Speeches”. *Legislative Studies Quarterly* 44(1): 97–131

Supplementary Reading:

- Watanabe, Kohei and Müller, Stefan (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io> (visited on 01/01/2024), Chapters 2–3
- Grimmer, Justin, Roberts, Margaret E., and Stewart, Brandon M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691207544/text-as-data>, Chapters 5 and 16 (instructor-provided excerpts)
- Muddiman, Ashley, McGregor, Shannon C., and Stroud, Natalie J. (2019). “(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries”. *Political Communication* 36(2): 214–226
- Hvitfeldt, Emil and Silge, Julia (2021). *Supervised Machine Learning for Text Analysis in R*. CRC Press. URL: <https://smltar.com/>, Chapters 2–5
- Manning, Christopher D., Raghavan, Prabhakar, and Schütze, Hinrich (2008). *Introduction to Information Retrieval*. Cambridge University Press. URL: <https://nlp.stanford.edu/IR-book/>, Chapters 2–3
- Ofosu, George K and Posner, Daniel N (2023). “Pre-analysis Plans: An Early Stocktaking”. *Perspectives on Politics* 21(1): 174–190
- Rathje, Steve et al. (2024). “GPT is an Effective Tool for Multilingual Psychological Text Analysis”. *Proceedings of the National Academy of Sciences of the United States of America (PNAS)* 121(34): e2308950121

- Jurafsky, Daniel and Martin, James H. (2026). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd. Online manuscript released August 19, 2026. URL: <https://web.stanford.edu/~jurafsky/slp3/>, Chapter 22

Technical Documents (not counted toward the reading limit):

- [quanteda documentation](#): consult the corpus, tokenisation, DFM, and dictionary examples used in the take-home lab

Take-home lab:

- Check the source, document unit, sampling boundaries, metadata, and access conditions of a supplied political-text corpus
- Follow the same example from corpus to tokens and a DFM using quanteda
- Apply a dictionary, normalize scores, and inspect ambiguous matches in context
- Compare a preprocessing choice and explain its effect on the substantive interpretation

Chinese segmentation with jieba or CKIP and extended dictionary comparisons are available as supplementary exercises. Students should use these checks to assess a potential project corpus before submitting their Week 7 abstract; they do not need to wait until Week 13 to begin obtaining data.

Week 5: National Day Observed (No Class)

No class on October 9.

Week 6: Human Coding and Document Classification

Topics:

- Classifying documents into pre-defined categories
- Creating training and test sets for document classification
- Assessing inter-coder reliability and annotation quality
- Evaluating classification performance with precision, recall, F1, and related metrics
- Bag-of-words classifiers and their relationship to transformer-based classifiers

Core Reading:

- Müller, Stefan (2022). “The Temporal Focus of Campaign Communication”. *The Journal of Politics* 84(1): 585–590
- Gilardi, Fabrizio, Alizadeh, Meysam, and Kubli, Maël (2023). “ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks”. *Proceedings of the National Academy of Sciences* 120(30): e2305016120
- Peterson, Andrew and Spirling, Arthur (2018). “Classification Accuracy as a Substantive Quantity of Interest: Measuring Polarization in Westminster Systems”. *Political Analysis* 26(1): 120–128

Supplementary Reading:

- Lacy, Stephen et al. (2015). “Issues and Best Practices in Content Analysis”. *Journalism & Mass Communication Quarterly* 92(4): 791–811
- Hvitfeldt, Emil and Silge, Julia (2021). *Supervised Machine Learning for Text Analysis in R*. CRC Press. URL: <https://smltar.com/>, Chapter 7
- Manning, Christopher D., Raghavan, Prabhakar, and Schütze, Hinrich (2008). *Introduction to Information Retrieval*. Cambridge University Press. URL: <https://nlp.stanford.edu/IR-book/>, Chapters 13, 14, and 15
- Hanretty, Chris (2023). *Replicating Mueller, ‘The Temporal Focus of Campaign Communication’*. Blog post demonstrating BERT fine-tuning replication. URL: <https://www.chrishanretty.co.uk/posts/finetuning/>
- Kapoor, Sayash and Narayanan, Arvind (2023). “Leakage and the Reproducibility Crisis in Machine-Learning-Based Science”. *Patterns* 4(9): 100804
- Peterson, Andrew and Spirling, Arthur (2018). “Classification Accuracy as a Substantive Quantity of Interest: Measuring Polarization in Westminster Systems”. *Political Analysis* 26(1): 120–128
- Liao, Yen-Chieh and Tang, Li (2025). “Electoral Systems and Geographically-Targeted Oversight: Evidence from Taiwan Legislative Yuan”. *Electoral Studies*
- Castanho Silva, Bruno, Pullan, Daniel, and Wäckerle, Jens (2024). “Blending In or Standing Out? Gendered Political Communication in 24 Democracies”. *American Journal of Political Science*

- Birkenmaier, Lukas, Lechner, Clemens M., and Wagner, Claudia (2024). “The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation”. *Communication Methods and Measures* 18(3): 249–277. DOI: [10.1080/19312458.2023.2285765](https://doi.org/10.1080/19312458.2023.2285765)
- Benoit, Kenneth, Conway, Drew, et al. (2016). “Crowd-sourced Text analysis: Reproducible and Agile Production of Political Data”. *American Political Science Review* 110(2): 278–295

Take-home lab:

- Supervised machine learning in quanteda
- Training Naive Bayes and SVM classifiers
- Creating train/test splits and cross-validation workflows
- Evaluating classification accuracy as a substantive quantity of interest

Week 7: Supervised, Unsupervised, and Semi-Supervised Scaling

Due: Individual research abstract, October 23, 2026. Submit a research paper title and an abstract of no more than 1,000 words.

Topics:

- Text scaling: estimating ideological positions from text
- Wordscores: supervised scaling method
- Wordfish: unsupervised scaling method
- Latent Semantic Scaling (LSS): semi-supervised approach
- Applications to party manifestos, parliamentary speeches, and comparative politics

Core Reading:

- Laver, Michael, Garry, John, and Benoit, Kenneth (2003). “Extracting Policy Positions from Political Texts Using Words as Data”. *American Political Science Review* 97(2): 311–331
- Slapin, Jonathan B. and Proksch, Sven-Oliver (2008). “A Scaling Model for Estimating Time-Series Party Positions from Texts”. *American Journal of Political Science* 52(3): 705–722
- Watanabe, Kohei (2021). “Latent Semantic Scaling: A Semisupervised Text Analysis Technique for New Domains and Languages”. *Communication Methods and Measures* 15(2): 81–102. DOI: [10.1080/19312458.2020.1832976](https://doi.org/10.1080/19312458.2020.1832976)

Supplementary Reading:

- Slapin, Jonathan B., Kirkland, Justin H., et al. (2018). “Ideology, Grandstanding, and Strategic Party Disloyalty in the British Parliament”. *American Political Science Review* 112(1): 15–30
- Le Mens, Gaël and Gallego, Aina (2025). “Positioning Political Texts with Large Language Models by Asking and Averaging”. *Political Analysis*
- Zollinger, Delia (2024). “Cleavage Identities in Voters’ Own Words: Harnessing Open-Ended Survey Responses”. *American Journal of Political Science* 68(1): 139–159
- Gessler, Theresa and Hunger, Sophia (2022). “How the Refugee Crisis and Radical Right Parties Shape Party Competition on Immigration”. *Political Science Research and Methods* 10(3): 524–544
- Lim, Linette and Dukalskis, Alexander (2025). “The Use of ‘So-called’ as a Propaganda Device in China”. *Journal of East Asian Studies*: 1–15
- Müller, Stefan, Brazys, Samuel, and Dukalskis, Alexander (2024). “Discourse Wars and ‘Mask Diplomacy’: China’s Global Image Management in Times of Crisis”. *Political Research Exchange* 6(1): 2337632

Technical Documents (not counted toward the reading limit):

- [LSX: Introduction to Latent Semantic Scaling](#)

Take-home lab:

- Implementing Wordscores for party manifesto analysis
- Estimating party positions with Wordfish
- Semi-supervised scaling with LSS
- Comparing scaling results across methods

Week 8: Topic Models and Their Variants

Topics:

- Discovering patterns and issue attention with topic models
- Latent Dirichlet Allocation (LDA) and structural topic models (STM)
- Assumptions, advantages, and caveats of topic modeling
- Keyword-assisted topic models
- Validating topics as measures

Core Reading:

- Grimmer, Justin, Roberts, Margaret E., and Stewart, Brandon M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691207544/text-as-data>, Chapter 13 (instructor-provided excerpt)
- Roberts, Margaret E. et al. (2014). “Structural Topic Models for Open-Ended Survey Responses”. *American Journal of Political Science* 58(4): 1064–1082
- Eshima, Shusei, Imai, Kosuke, and Sasaki, Tomoya (2024). “Keyword-Assisted Topic Models”. *American Journal of Political Science* 68(2): 730–750

Supplementary Reading:

- Gessler, Theresa (2022). “Topic Models”. In: *Elgar Encyclopedia of Technology and Politics*. Ed. by Ceron, Andrea. Cheltenham: Edward Elgar Publishing: 108–111 (instructor-provided chapter excerpt)
- Ying, Luwei, Montgomery, Jacob M., and Stewart, Brandon M. (2022). “Topics, Concepts, and Measurement: A Crowdsourced Procedure for Validating Topics as Measures”. *Political Analysis* 30(4): 570–589
- Müller, Stefan, Kennedy, Garrett, and Maher, Tomás (2023). “Reactions to Experts in Deliberative Democracy: the 2016–2018 Irish Citizens’ Assembly”. *Irish Political Studies* 38(4): 467–488
- Catalinac, Amy (2016). “From Pork to Policy: The Rise of Programmatic Campaigning in Japanese Elections”. *The Journal of Politics* 78(1): 1–18
- Grootendorst, Maarten (2022). *BERTopic: Neural Topic Modeling with a Class-based TF-IDF Procedure*. arXiv preprint arXiv:2203.05794

Take-home lab:

- Implementing LDA topic models
- Comparing unseeded and keyword-assisted models
- Selecting and comparing candidate numbers of topics
- Validating and interpreting topic model results with example documents

Embedding-based topic models such as BERTopic are previewed here, with representation concepts developed further in Week 9.

Week 9: Describing and Comparing Texts: Readability, Similarity, and Word Embeddings

Topics:

- Readability and complexity as measures of political language
- Similarity, distance, keyness, and the choice of a comparison group
- From sparse word counts to Word2Vec, GloVe, and FastText representations
- Context-specific embeddings and embedding regression for comparing meaning across actors and groups
- Validation, bias, and multilingual applications in political and social-science research

Core Reading:

- Bischof, Daniel and Senninger, Roman (2018). “Simple Politics for the People? Complexity in Campaign Messages and Political Knowledge”. *European Journal of Political Research* 57(2): 473–495
- Rodriguez, Pedro L and Spirling, Arthur (2022). “Word Embeddings: What Works, What Doesn’ t, and How to Tell the Difference for Applied Research”. *Journal of Politics* 84(1): 101–115
- Rodriguez, Pedro L, Spirling, Arthur, and Stewart, Brandon M (2023a). “Embedding Regression: Models for Context-Specific Description and Inference”. *American Political Science Review* 117(4): 1255–1274

Supplementary Reading:

- Benoit, Kenneth, Munger, Kevin, and Spirling, Arthur (2019). “Measuring and Explaining Political Sophistication Through Textual Complexity”. *American Journal of Political Science* 63(2): 491–508
- Grimmer, Justin, Roberts, Margaret E., and Stewart, Brandon M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691207544/text-as-data>, Chapter 8 (instructor-provided excerpt)
- Hengel, Erin (2022). “Publishing While Female: Are Women Held to Higher Standards? Evidence from Peer Review”. *The Economic Journal* 132(648): 2951–2991
- Shulman, H. C., Markowitz, D. M., and Rogers, T. (2024). “Reading Dies in Complexity: Online News Consumers Prefer Simple Writing”. *Science Advances* 10(23): eadn2555

- Blumenau, Jack (2021). “The Effects of Female Leadership on Women’s Voice in Political Debate”. *British Journal of Political Science* 51(2): 750–771
- Wirsching, Elisa M et al. (2025). “Multilanguage Word Embeddings for Social Scientists: Estimation, Inference, and Validation Resources for 157 Languages”. *Political Analysis* 33(2): 156–163
- Reimers, Nils and Gurevych, Iryna (2019). “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics: 3980–3990
- Rodriguez, Pedro L., Spirling, Arthur, and Stewart, Brandon M. (2023b). *Replication Data for: Embedding Regression: Models for Context-Specific Description and Inference*. Harvard Dataverse. Version V1. URL: <https://doi.org/10.7910/DVN/NKETXF>
- Holman, Mirya R., Johnson, Rachel, and Simko, Tyler (2024). *Measuring Conflict in Local Politics*. OSF PrePrint. URL: <https://osf.io/vst9g>
- Mikolov, Tomas et al. (2013). “Efficient Estimation of Word Representations in Vector Space”. In: *Proceedings of the International Conference on Learning Representations (ICLR)*
- Pennington, Jeffrey, Socher, Richard, and Manning, Christopher D. (2014). “GloVe: Global Vectors for Word Representation”. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Doha, Qatar: Association for Computational Linguistics: 1532–1543
- Bojanowski, Piotr et al. (2017). “Enriching Word Vectors with Subword Information”. *Transactions of the Association for Computational Linguistics* 5: 135–146

Take-home lab:

- Compare readability or count-based similarity for a small set of political texts
- Use a supplied embedding model to inspect semantic neighbors or document similarity
- Return to example passages to assess what the differences mean substantively

Extended embedding-regression and multilingual workflows are provided as supplementary exercises.

Week 10: NLP Pipelines and Transformers: Annotation, Classification, and Interpretability

Topics:

- What POS tagging, named entity recognition, and dependency parsing extract from political texts

- The intuition behind contextual representations, attention, and BERT-style classification
- Published social-science applications: when transformers improve on bag-of-words baselines
- Validation and error analysis, with a pre-run illustration of fine-tuning and model interpretation

Core Reading:

- Kroon, Anne et al. (2024). “Advancing Automated Content Analysis for a New Era of Media Effects Research: The Key Role of Transfer Learning”. *Communication Methods and Measures* 18(2): 142–162
- Timoneda, Joan C and Vallejo Vera, Sebastián (2025). “BERT, RoBERTa, or DeBERTa? Comparing Performance Across Transformers Models in Political Science Text”. *The Journal of Politics* 87(1): 347–364. DOI: [10.1086/730737](https://doi.org/10.1086/730737)
- Laurer, Moritz et al. (2024). “Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI”. *Political Analysis* 32(1): 84–100

Supplementary Reading:

- Straka, Milan, Hajic, Jan, and Straková, Jana (2016). “UDPipe: Trainable Pipeline for Processing CoNLL-U files Performing Tokenization, Morphological Analysis, POS tagging and parsing”. In: *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC’16)*: 4290–4297
- Akbik, Alan et al. (2019). “FLAIR: An Easy-to-use Framework for State-of-the-art NLP”. in: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (demonstrations)*: 54–59
- Vaswani, Ashish et al. (2017). “Attention is All You Need”. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*: 5998–6008
- Jurafsky, Daniel and Martin, James H. (2026). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd. Online manuscript released August 19, 2026. URL: <https://web.stanford.edu/~jurafsky/slp3/>, relevant NLP, transformer, and pretraining chapters
- Tunstall, Lewis, Werra, Leandro von, and Wolf, Thomas (2022). *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*. O’Reilly Media. URL: <https://www.oreilly.com/library/view/natural-language-processing/9781098136789/>, Chapter 4 (instructor-provided excerpt)

- Devlin, Jacob et al. (2019). “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. Association for Computational Linguistics: 4171–4186
- Liu, Yinhan et al. (2019). “RoBERTa: A Robustly Optimized BERT Pretraining Approach”. *arXiv preprint arXiv:1907.11692*
- Sanh, Victor et al. (2019a). “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter”. In: *5th Workshop on Energy Efficient Machine Learning and Cognitive Computing at NeurIPS 2019*
- Bucher, Matthias J. J. and Martini, Moritz (2024). *Fine-Tuned ‘Small’ LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification*. *arXiv preprint arXiv:2406.08660*. URL: <https://arxiv.org/abs/2406.08660>
- Sui, Yuan (Cecilia) (2025). “High Accuracy with Low Costs: The Pretrain-Finetune Paradigm for Classification with Transformer-based Language Models”. Working paper. URL: https://osf.io/preprints/socarxiv/w7q43_v1
- Müller, Stefan and Proksch, Sven-Oliver (2024). “Nostalgia in European Party Politics: A Text-Based Measurement Approach”. *British Journal of Political Science* 54(3): 993–1005
- Widmann, Tobias and Wich, Maximilian (2024). “Creating and Comparing Dictionary, Word Embedding, and Transformer-Based Models to Measure Discrete Emotions in German Political Text”. *Political Analysis* 32(1): 1–16
- Ribeiro, Marco Tulio, Singh, Sameer, and Guestrin, Carlos (2016). “‘Why Should I Trust You?’: Explaining the Predictions of Any Classifier”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. San Francisco, California: ACM: 1135–1144
- Murdoch, W. James et al. (2019). “Definitions, Methods, and Applications in Interpretable Machine Learning”. *Proceedings of the National Academy of Sciences* 116(44): 22071–22080
- Lundberg, Scott M. and Lee, Su-In (2017). “A Unified Approach to Interpreting Model Predictions”. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*: 4765–4774
- Belinkov, Yonatan (2022). “Probing Classifiers: Promises, Shortcomings, and Advances”. *Computational Linguistics* 48(1): 207–219

Technical Documents (not counted toward the reading limit):

- Hugging Face (2023). *Transformers: State-of-the-Art Natural Language Processing*. Documentation and tutorials. URL: <https://huggingface.co/docs/transformers/index> (Skim relevant tutorials)
- The Illustrated Transformer: <https://jalammar.github.io/illustrated-transformer/>
- Captum Tutorial: <https://captum.ai/tutorials/>
- udpipe R Documentation: <https://bnosac.github.io/udpipe/en/>
- flaiR R Documentation: <https://davidycliao.github.io/flaiR/>

Take-home lab:

- Inspect POS tags or named entities in a short legislative or news-text example
- Run a supplied pre-trained classification workflow and compare its errors with a classical baseline
- Examine one explanation or error case and distinguish model behavior from substantive evidence

The session prioritizes concepts, social-science applications, and validation rather than implementation of every NLP task. Detailed architecture, full fine-tuning, SHAP/LIME implementations, and extended Chinese-language pipelines are supplementary. The take-home lab uses supplied models; students are not required to train a transformer or implement all interpretation tools.

Week 11: NTU Sports Day (No Class)

No class on November 20.

Week 12: Large Language Models and Multi-Agent Systems

Topics:

- What LLMs can do well for text analysis
- Zero-shot and few-shot classification, summarisation, and annotation
- Commercial models vs. open-source models
- Robust and reproducible research with LLMs
- LLM-based measurement of party positions and validation against human and expert judgments
- Multi-agent workflows and the limitations of political or social-science simulations

- Trade-offs between classical methods, transformers, and generative models

Core Reading:

- Bail, Christopher A. (2024). “Can Generative AI Improve Social Science?” *Proceedings of the National Academy of Sciences* 121(21): e2314021121
- Benoit, Kenneth, De Marchi, Scott, et al. (2025). “Using Large Language Models to Analyze Political Texts through Natural Language Understanding”. *American Journal of Political Science*. Advance online publication: 1–17. DOI: [10.1111/ajps.70050](https://doi.org/10.1111/ajps.70050)
- Laurer, Moritz et al. (2025). “On Measurement Validity and Language Models: Increasing Validity and Decreasing Bias with Instructions”. *Communication Methods and Measures* 19(1): 46–62. DOI: [10.1080/19312458.2024.2378690](https://doi.org/10.1080/19312458.2024.2378690)

Supplementary Reading:

- Cova, Jade and Schmitz, Lukas (2024). *A Primer for the Use of Classifier and Generative Large Language Models in Social Science Research*. OSF PrePrint. URL: <https://doi.org/10.31219/osf.io/r3qng>
- Ornstein, Joseph T., Blasingame, Elise N., and Truscott, Jake S. (2025). “How to Train Your Stochastic Parrot: Large Language Models for Political Texts”. *Political Science Research and Methods*
- Spirling, Arthur (2023). “Why Open-source Generative AI Models Are an Ethical Way Forward for Science”. *Nature* 616(7957): 413–413
- Ollion, Étienne et al. (2024). “The Dangers of Using Proprietary LLMs for Research”. *Nature Machine Intelligence* 6(1): 4–5
- Hu, Tiancheng et al. (2025). “Generative Language Models Exhibit Social Identity Biases”. *Nature Computational Science* 5: 65–75
- Brown, Tom B. et al. (2020). “Language Models are Few-Shot Learners”. *Advances in Neural Information Processing Systems* 33: 1877–1901
- Wei, Jason et al. (2022). “Emergent Abilities of Large Language Models”. *Transactions on Machine Learning Research*
- Kojima, Takeshi et al. (2022). “Large Language Models are Zero-Shot Reasoners”. *Advances in Neural Information Processing Systems* 35: 22199–22213
- Ziems, Caleb et al. (2024). “Can Large Language Models Transform Computational Social Science?” *Computational Linguistics* 50(1): 237–291

- Park, Joon Sung et al. (2023). “Generative Agents: Interactive Simulacra of Human Behavior”. In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM: 1–22
- Argyle, Lisa P. et al. (2023). “Out of One, Many: Using Language Models to Simulate Human Samples”. *Political Analysis* 31(3): 337–351
- Bisbee, James et al. (2024). “Synthetic Replacements for Human Survey Data? The Perils of Large Language Models”. *Political Analysis* 32(4): 401–416
- Ollion, Étienne et al. (2023). *ChatGPT for Text Annotation? Mind the Hype!* SocArXiv preprint. URL: <https://doi.org/10.31235/osf.io/x58kn>
- Egami, Naoki, Hinck, Musashi, et al. (2023). “Using Imperfect Surrogates for Downstream Inference: Design-based Supervised Learning for Social Science Applications of Large Language Models”. In: *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*: 68589–68601
- Liao, Yen-Chieh, Lim, Linette, and Jankin, Slava (2025). “Multi-Agent LLM Systems for Synthetic Survey Experiments in Ethically Constrained Settings”. Working Paper. URL: https://www.dropbox.com/scl/fi/pnpv3t1dqji7qxnuoihqx/Multi_Agent_Systems_with_LLMs.pdf?rlkey=du7hl5mainlk6lp0a0ki6b5f0&e=1&dl=0
- Wu, Qingyun et al. (2023). “AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation”. *arXiv preprint arXiv:2308.08155*

Technical Documents (not counted toward the reading limit):

- AutoGen Documentation: <https://microsoft.github.io/autogen/>

Take-home lab:

- Running LLMs through APIs and local models
- Designing and validating prompts for political text analysis
- Comparing LLM outputs with classical and transformer-based approaches
- Documenting model use for reproducible research

A simple multi-agent demonstration is available as supplementary material.

Week 13: New Forms of Textual Data: Multilingual and Multimodal Sources

Due: Midterm research design report, December 4, 2026. Submit a research design report of no more than 10 pages and 2,500 words.

Students should arrive with a draft based on the data-feasibility checks introduced in Weeks 2 and 4. They may use insights from the session to refine their treatment of data sources and measurement before the 23:59 submission deadline.

Topics:

- New research opportunities in multilingual news, parliamentary recordings, interviews, and campaign videos
- Multilingual analysis: separate-language models, input alignment through translation or shared representations, and cross-language anchors
- Speech-to-text and OCR: turning spoken or scanned material into analyzable text while checking names, policy terms, and negation
- Multimodal sources: what transcripts and captions preserve, and what they omit about voice, images, and context
- Transformation, annotation, and interpretation: locating model-generated data within the research workflow
- Comparing original and transformed texts to assess whether errors change sentiment, issue attention, or substantive conclusions
- Ethical access, source provenance, and transparent reporting of transcription, translation, and annotation choices

Core Reading:

- Proksch, Sven-Oliver, Wratil, Christopher, and Wäckerle, Jens (2019). “Testing the Validity of Automatic Speech Recognition for Political Text Analysis”. *Political Analysis* 27(3): 339–359. doi: [10.1017/pan.2018.62](https://doi.org/10.1017/pan.2018.62)
- Licht, Hauke and Lind, Fabienne (2023). “Going Cross-Lingual: A Guide to Multilingual Text Analysis”. *Computational Communication Research* 5(2): 1–31. doi: [10.5117/CCR2023.2.2.LICH](https://doi.org/10.5117/CCR2023.2.2.LICH)
- Birkenmaier, Lukas, Lechner, Clemens M., and Wagner, Claudia (2024). “The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation”. *Communication Methods and Measures* 18(3): 249–277. doi: [10.1080/19312458.2023.2285765](https://doi.org/10.1080/19312458.2023.2285765)

Supplementary Reading:

- Merz, Nicolas, Regel, Sven, and Lewandowski, Jirka (2016). “The Manifesto Corpus: A New Resource for Research on Political Parties and Quantitative Text Analysis”. *Research & Politics* 3(2): 1–8
- Wickham, Hadley, Çetinkaya-Rundel, Mine, and Grolemund, Garrett (2023). *R for Data Science (2nd Edition)*. O’Reilly Media. URL: <https://r4ds.hadley.nz/>, Chapters 5–7
- Bauer, Paul C. and Landesvatter, Camille (2023). *APIs for Social Scientists: A Collaborative Review*. Online Collaborative Book. URL: https://paulcbauer.github.io/apis_for_social_scientists_a_review/ (Skim potentially relevant chapters)
- Barari, Soubhik and Simko, Tyler (2023). “LocalView, a Database of Public Meetings for the Study of Local Politics and Policy-making in the United States”. *Scientific Data* 10: 135
- Fiesler, Casey, Beard, Nathan, and Keegan, Brian C. (2020). “No Robots, Spiders, or Scrapers: Legal and Ethical Regulation of Data Collection Methods in Social Media Terms of Service”. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 14: 187–196
- Salganik, Matthew J. (2017). *Bit by Bit: Social Research in the Digital Age*. Princeton: Princeton University Press. URL: <https://www.bitbybitbook.com/en/1st-ed/preface/>, Chapter 6
- Stoltz, Dustin S, Taylor, Marshall A, and Dudley, Jennifer SK (2023). “A Tool Kit for Relation Induction in Text Analysis”. *Sociological Methods & Research*: 00491241241233242
- Lim, Linette and Dukalskis, Alexander (2025). “The Use of ‘So-called’ as a Propaganda Device in China”. *Journal of East Asian Studies*: 1–15
- Müller, Stefan, Brazys, Samuel, and Dukalskis, Alexander (2024). “Discourse Wars and ‘Mask Diplomacy’: China’s Global Image Management in Times of Crisis”. *Political Research Exchange* 6(1): 2337632
- Liao, Yen-Chieh, Zheng, Chen, et al. (2025). “Political Text Analysis with Embedding Regression: From Multilingual to Cross-lingual Application”. Working Paper

Technical Documents (not counted toward the reading limit):

- MarkItDown Documentation: <https://pypi.org/project/markitdown/>

Take-home lab:

- Inspect a short political speech excerpt and a reference transcript or translation
- Compare it with a supplied machine-transcribed or machine-translated version; identify errors in names, negation, or policy terms
- Run a supplied dictionary or keyword comparison on both versions and inspect any differences in the resulting measure
- Consider whether the differences change the substantive interpretation and what should be documented

The instructor will provide prepared inputs and code, so the exercise does not require model training or live API access. Advanced retrieval and file-conversion workflows remain optional technical resources; the emphasis is on how new data sources enable research and how transformations can affect the evidence.

Week 14: Poster Workshop: Format, Visual Presentation, and 5-Minute Talk

Poster Workshop Format:

- Focus on poster format, visual hierarchy, figure/table placement, and readable text density
- Practice the structure of a 5-minute presentation: research question, data, method, key finding, and implication
- Discuss concrete revision directions collectively before the formal Week 15 poster session
- Students are encouraged to share a poster draft through Overleaf before the formal presentation and use after-class discussions or office hours to refine the research content, visual presentation, and brief oral introduction for visitors

Recommended Resources:

- PolMeth Poster Guidelines: <https://polmeth.emory.edu/instructions.html>
- PolMeth Poster Examples: [PolMeth 2025 Program](#); see the examples linked under *Final Poster Presentation (25%)* below

Week 15: Final Poster Presentations

Formal assessed presentation: December 18, 2026 (Week 15). Faculty members from the department or institute will be invited to attend.

Poster Session Format:

- Present the research question, data and methods, key findings, and implications through clear, concise text, figures, and tables
- Give an approximately 5-minute introduction, followed by questions and feedback from faculty, classmates, and other participants
- Remain beside the poster throughout the display period to introduce the research to visitors and answer questions

Week 16: Constitution Day Holiday (No Class)

No class on December 25 because of the Constitution Day holiday. Students revise their projects independently after the Week 15 poster session. **Manuscript due: TBA. Bonus claim deadline: TBA.** Both deadlines will be announced on NTU COOL. Submit the final research paper through Overleaf with instructor editing access. Bonus claims are also submitted through a shared Overleaf project and must include the manuscript, submission confirmation, and required acceptance or editorial decision letter.

References

- Akbik, Alan et al. (2019). “FLAIR: An Easy-to-use Framework for State-of-the-art NLP”. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (demonstrations)*: 54–59.
- Alexander, Rohan (2023). *Telling Stories with Data: With Applications in R*. New York: CRC Press. URL: <https://tellingstorieswithdata.com>.
- Argyle, Lisa P. et al. (2023). “Out of One, Many: Using Language Models to Simulate Human Samples”. *Political Analysis* 31(3): 337–351.
- Baden, Christian et al. (2022). “Three Gaps in Computational Text Analysis Methods for Social Sciences: A Research Agenda”. *Communication Methods and Measures* 16(1): 1–18. DOI: [10.1080/19312458.2021.2015574](https://doi.org/10.1080/19312458.2021.2015574).
- Bail, Christopher A. (2024). “Can Generative AI Improve Social Science?” *Proceedings of the National Academy of Sciences* 121(21): e2314021121.
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8 Grading

8.1 Assessment and Grading

Assessment follows the development of an original research project, beginning with an individual research paper abstract (10%), due in Week 7. The midterm research design report (20%), due in Week 13, and final research paper (25%), with its deadline to be announced, are individual submissions in which students develop their own original research using the course's methods. The final project is presented in a formal poster session in Week 15 (25%). Week 14 is a poster workshop that prepares students for the formal presentation and counts toward class participation. Class participation accounts for 20% and includes discussion of readings, methods, and applications, peer feedback, and contributions to the course discussion board. The co-authorship rules for optional conference or journal bonus submissions do not change the individual-submission requirement for assessed coursework.

Submission deadlines: Written assignments and bonus claims are due at 23:59 (Taiwan time, UTC+8) on the dates stated below or subsequently announced by the instructor. Written coursework and bonus claims are submitted through shared Overleaf projects. The final research paper and bonus-claim deadlines are TBA. Poster presentations take place during the scheduled Week 15 class. Approved extensions are handled under the Late Submission and Student Leave policy.

Submitting through Overleaf: Prepare research abstracts, research design reports, final papers, and posters in **L^AT_EX** using the supplied templates. For each assessed submission, use a separate Overleaf project (<https://www.overleaf.com>) named with your name and the assignment. By the deadline, invite davidycliao@gmail.com with **Editor (can edit)** access so the instructor can provide comments and edit directly. A free account allows one invited collaborator per project and is sufficient for this arrangement; no paid subscription is required. View-only access is not sufficient for submission.

The shared project must contain the complete, current source files and compile successfully to PDF. Local **L^AT_EX** editing is permitted, but upload the current files to Overleaf before submission. After the deadline, wait for instructor feedback or approval before making substantive revisions. Quarto may be used for analysis notebooks, but not as an alternative authoring format for written submissions or posters.

Component	Weight	Deadline
Research Paper Abstract	10%	Oct 23 (Week 7)
Midterm Research Design Report	20%	Dec 4 (Week 13)
Class Participation	20%	Throughout the semester
Final Poster Presentation	25%	Dec 18 (Week 15), in class
Final Research Paper	25%	TBA
Total	100%	

Research Paper Abstract (10%)

Students submit a proposed research paper title and an abstract of no more than 1,000 words. The abstract introduces the study they plan to develop in the midterm and final research papers.

Due: Week 7, October 23, 2026. This is an individual submission.

Midterm Research Design Report (20%)

Students apply methods learned in the course to their own research project or thesis. The midterm paper sets up the study that the final research project will complete, so it should establish the research question, situate it in the literature, and lay out the research design.

Requirements:

- Abstract
- Introduction: research question and theoretical motivation
- Literature review situating the study
- Research design: data, methods, and analytical strategy
- Preliminary results or planned analytical strategy
- Feasibility, expected findings, and contribution
- Manuscript: up to 10 pages, no more than 2,500 words
- Due: Week 13, December 4, 2026

Class Participation (20%)

Active engagement in the following activities:

- Class discussions of readings, methods, and applications
- Peer feedback activities
- Constructive questions and answers on the NTU COOL course discussion board or course-code GitHub Issues

Final Poster Presentation (25%)

Students are encouraged to share a poster draft with the instructor through Overleaf before the formal presentation. They are also encouraged to discuss their drafts after class or during office hours, focusing on poster format, visual presentation, and how to give visitors a brief, clear oral introduction to the research. These discussions should help students refine both their research content and its presentation. Week 14 provides a preparatory poster workshop.

All formally assessed final poster presentations will take place in Week 15 (December 18, 2026). Students should refer to the PolMeth poster guidelines when preparing their posters (<https://polmeth.emory.edu/instructions.html>).* Posters should use clear, concise visual design to present the research question, data and methods, key findings, and their substantive implications, using figures and tables to communicate the analysis and results.

Faculty members from the department or institute will be invited to the Week 15 session. Each student will first give an approximately 5-minute introduction to their poster, followed by questions and feedback from faculty, classmates, and other participants. During the poster display period, students should remain beside their posters to introduce their research to visitors and answer questions.

Final Research Paper (25%)

Original text analysis research project demonstrating mastery of course methods, written as a Research Note in the style of *Political Science Research and Methods* (PSRM).

Requirements:

- Build upon and complete the midterm paper with full empirical analysis
- Strengthened research design with robust inference and interpretation
- Rigorous analysis with proper validation and sensitivity tests
- Discussion of substantive findings, limitations, and contributions

*For examples, see the [PolMeth 2025 Program](#) and the following posters from the 2025 PolMeth Annual Meeting: Soyeon Jeon; Yen-Chieh Liao; Zikai Li; and Yuki Atsusaka, Diana Da In Lee, and Hane Lee.

- Main text: up to 4,000 words for this course, including footnotes and text in main-document tables and figure captions, but excluding references and a separate supplementary appendix. The assignment is inspired by the PSRM Research Note format; this is a course-specific word-count rule, not an exact reproduction of the journal's submission requirements. PSRM's own 4,000-word limit includes references and other main-manuscript material. Students preparing an actual journal submission must check the journal's [current author instructions](#).
- Appendix: supplementary material with no word or page limit for this course. Students are encouraged to use the appendix to provide as much additional analysis as is useful, including robustness checks, alternative specifications, validation exercises, additional tables and figures, annotation protocols, prompts, model details, and code documentation.
- Reproducible code repository via GitHub or CodeOcean (<https://codeocean.com>)
- Manuscript due: TBA (to be announced on NTU COOL)

Suggested Topics:

- Taiwan legislative behavior: roll call votes, speeches, and parliamentary questions
- Party manifesto analysis across countries or time
- Legislative speech analysis
- Social media political discourse analysis
- News media framing of political issues
- Comparative analysis of political rhetoric
- Analysis of CCP wolf warrior diplomacy and propaganda

8.2 Bonus Opportunities (Up to 15% Extra Credit)

Conference Acceptance Bonus: Up to 5% extra credit.

An eligible acceptance must be documented by the course bonus-claim deadline below. The following are examples of venues, not a list of deadlines for this course; students must check each venue's current call for papers. Submissions whose decisions arrive after the course cutoff may still support students' research development but are not eligible for this semester's bonus.

- Japanese Society for Quantitative Political Science (JSQPS) (<https://sites.google.com/view/jsqps/>)
- European Political Science Society (EPSS) (<https://epssnet.org>)
- Midwest Political Science Association (MPSA) (<https://www.mpsanet.org>)

- American Political Science Association (APSA) (<https://www.apsanet.org>)
- PolMeth Conferences:
 - PolMeth Annual Meeting (<https://polmeth.org/spm-central>)
 - Europe PolMeth (<https://polmeth.eu>)
 - Asian PolMeth (<https://polmeth.org/asia>)
- COMPTExT Conference (<https://www.comptextconference.org>)
- Taiwan Political Science Association (TPSA) (<https://www.tpsa.org.tw>)

Journal Submission Bonus: Up to 10% extra credit

- Acceptance, Revise & Resubmit, or Major/Minor Revision: +10%
- Reject or Desk Reject: +5%
- Eligible journals: SSCI-indexed political science or social science journals

Bonus Policy:

- **Bonus claim deadline: TBA.** The cutoff date will be announced on NTU COOL. Share the manuscript with the instructor through an Overleaf project with Editor access, and upload the submission confirmation and the acceptance or editorial decision letter required for the claimed bonus to the same project. The decision must have been received and all supporting documents made available by the announced deadline. This is the course's bonus-claim cutoff, not a conference or journal submission deadline.
- A submission still awaiting a decision at the cutoff does not receive provisional bonus credit. Decisions received after the cutoff are not counted toward this semester's course bonus. A conference submission without acceptance does not qualify for the conference acceptance bonus. The listed journal decision categories determine the journal bonus.
- Submissions used to claim bonus credit must comply with NTU regulations on academic integrity, research ethics, student conduct, authorship, copyright, data protection, and generative AI use, as well as the rules of the relevant conference or journal. A submission that violates these rules is not eligible for bonus credit and may be handled under the relevant NTU procedures.
- Students must use accurate institutional and geographic identification in the manuscript text, submission metadata, cover letter, acknowledgements, slides, posters, and any public-facing materials. When listing affiliation, students should use the official name *National Taiwan University*. This follows NTU's [official public name and address usage](#) and the [university record](#) asking members to maintain the official English name *National Taiwan University*. Students

must not alter, omit, or downgrade the university name, and must not use wording that misrepresents or diminishes Taiwan's status. Where a location is required, use a form consistent with NTU's official usage, such as "Taipei, Taiwan" or "Taipei, Taiwan (R.O.C.)". If a conference, journal, submission system, or editor requests wording that changes or downgrades NTU or Taiwan, students must consult the instructor before proceeding.

- This rule does not prohibit evidence-based criticism of Taiwanese politics, institutions, or public policy. It applies to accurate affiliation, geographic description, professional conduct, and non-misleading representation in scholarly submissions.
- Students writing an MA thesis are encouraged to develop thesis-related conference or journal submissions in co-authorship with their thesis supervisor, where appropriate and consistent with NTU academic-integrity, authorship, and research-ethics regulations. For students in the Graduate Institute of National Development, eligible thesis-related co-authored submissions for course bonus credit must *be co-authored with a faculty member affiliated with the Graduate Institute of National Development*. To claim course bonus credit for a thesis-related co-authored submission, the student must be listed as both first author and corresponding author.
- If a student achieves both conference and journal bonuses, both can be claimed (maximum 15% total)
- Students are strongly encouraged to discuss their submission plans with the instructor before submitting

9 Course Policies

9.1 Academic Integrity and Plagiarism

Although this should be obvious, plagiarism is not allowed. Students must acknowledge all sources of ideas, text, code, data, figures, tables, and research materials that are not their own. This includes direct copying without quotation and citation, close paraphrasing without citation, reusing one's own previous work without acknowledgement, unauthorized collaboration, fabricated data or references, and submitting AI-generated material as one's own work without disclosure when disclosure is required.

This course follows National Taiwan University policies on academic integrity and student conduct. Relevant NTU resources include the Office of Research Integrity (<https://ori.ntu.edu.tw/>), the NTU academic ethics regulations page (<https://ori.ntu.edu.tw/laws>), the NTU student academic ethics page (https://ori.ntu.edu.tw/ethics/index/ethics_type/2), and the NTU student regulations page maintained by the Office of Student Affairs (<https://advisory.ntu.edu.tw/CMS/Page/265>). Suspected academic misconduct will be handled according to NTU procedures and may receive a zero on the relevant course component. Serious or repeated cases may be referred to the department, college, or relevant university office for formal handling under NTU regulations.

Similarity-checking tools may be used as an aid, but final judgments require human review of the evidence and context.

9.2 Use of Generative AI

This course follows NTU's *Guidelines for the Use of Generative AI and Research Integrity* (<https://ori.ntu.edu.tw/news/download/fileSn/59>). Generative AI tools may be used as support tools, but they do not replace students' own academic judgment, original analysis, source verification, or responsibility for the final submission. Generative AI tools must not be listed as authors, and students must not submit unverified AI-generated citations, data, code, analysis, translations, or prose as their own work. Students should retain relevant records of AI use when the tool materially shapes research design, analysis, coding, writing, or revision, and they must not enter personal data, sensitive data, unpublished research materials, confidential review materials, or other protected information into open generative AI systems unless this is lawful, necessary, and appropriately protected.

Students may use generative AI tools for the following aspects of their assignments:

- Conceptualisation: brainstorming topics, outlining arguments, planning research designs, and organizing study schedules.
- Initial literature screening: identifying potentially relevant literature, search terms, debates, or venues for further human verification.
- Supporting writing and editing: improving clarity, structure, grammar, translation, or formatting, provided that the intellectual argument and substantive analysis remain the student's own.
- Audiovisual and interactive components, where relevant to an assignment, such as help with captions, storyboards, interface drafts, or presentation materials.

Students must always disclose their use of generative AI in every assignment, poster, final paper, and bonus-credit submission. To draft this declaration, students may use the free, open-source AI Disclosure Generator at <https://disclose-ai.app>. The declaration does not need to include every exact prompt, but it must state which tool and version were used, what the student asked the tool to do, how the output was used, and how the student checked or validated the output. If no generative AI tools were used, students must include the following statement: "No generative AI tools were used in the preparation of this assignment."

Assignments submitted without an AI-use declaration will not be accepted until a complete declaration is provided. Misrepresenting AI use, failing to verify AI-generated content, fabricating sources or data through AI, or using AI to bypass the assignment's learning goals may be treated as academic misconduct under the relevant NTU procedures.

9.3 Late Submission and Student Leave

Students should submit all assignments by the stated deadline. Students who cannot attend class, participate in a presentation, take an in-class assessment, or submit coursework on time because of illness, emergency, official duty, mental-health leave, bereavement, childbirth, military service, or another ground covered by NTU regulations must follow the National Taiwan University student leave procedures (<https://advisory.ntu.edu.tw/CMS/Page/44>). Students should notify the instructor as early as possible and, where applicable, complete the leave request through the NTU student leave system. In urgent situations, students should first notify the instructor by email or another available channel, then complete the required NTU leave procedure afterward within the university's time limits.

Approved leave or an instructor-approved extension does not automatically waive coursework. The instructor will arrange a reasonable make-up deadline or alternative activity when appropriate. Students who do not follow NTU leave procedures and do not receive prior approval will be marked absent for missed in-class activities. Coursework submitted after the deadline without approved NTU leave or an instructor-approved extension may receive no credit; partial credit for late work, if any, is at the instructor's discretion and will be applied consistently across students.

9.4 Syllabus Modification Rights

The instructor reserves the right to reasonably alter elements of the syllabus during the semester. The reading list may be adjusted to keep pace with new research, software changes, student interests, and the actual pace of the course. Specific session content may also change depending on participants' prior knowledge and research projects. Any substantive changes will be announced by email and through the course platform or GitHub repository.

9.5 Dignity and Respect

National Taiwan University is committed to a respectful, inclusive, and safe learning environment. Students have the right to study in an environment free from bullying, harassment, discrimination, sexual harassment, sexual assault, sexual bullying, stalking, and other conduct that undermines dignity or safety. Classroom discussion may involve politically sensitive materials, but disagreement should remain evidence-based, professional, and respectful.

Students who experience or witness harassment, gender-based misconduct, discrimination, bullying, or threats to their safety are encouraged to seek support. Reports and requests for assistance will be handled in accordance with the relevant NTU procedures. Students are also welcome to contact me directly for support and guidance.

NTU resources include the [Gender Equity Education Committee](#) (國立臺灣大學性別平等教育委員會), the [NTU sexual harassment prevention resources](#) (國立臺灣大學性騷擾防治專區), the NTU sexual harassment prevention contact line (02) 3366-5903 and email leader@ntu.edu.tw, and the Gender Equity Education Committee complaint lines (02) 3366-9607 and (02) 3366-9608 and

email gender@ntu.edu.tw. For immediate safety concerns, contact the Campus Security Center at **(02) 3366-9119**. If someone's life or physical safety is in danger, call **119** or **110**.